11 · Patterns of inheritance

The words of this track, as it says them. Bold brown is the examiner.

Patterns of inheritance. Four marks.

Briefly explain these patterns of inheritance, with an example of each: codominance, incomplete dominance, polygenic inheritance, and mitochondrial inheritance.

Four patterns, four marks, and an example for each. Her 2023 revision deck puts them in a four-column table, and that's the shape to write it in.

The hook: co means both show. Incomplete means a blend. Poly means many genes. Mito means mum only.

Codominance. Both alleles are fully expressed in the heterozygote, each making its own protein. The example is the A B O blood groups: I A and I B together give group A B, with both the A and the B antigens on the red cells.

Incomplete dominance. One allele doesn't completely mask the other, so the heterozygote is somewhere in between. Her example is sickle cell. The heterozygote has both normal and sickle-shaped cells, with about a third of the haemoglobin abnormal, and more resistance to malaria.

Polygenic inheritance. Many genes control one trait, so you get a continuous range, a bell curve, rather than neat categories. Examples: height, skin colour, and eye colour.

Mitochondrial inheritance. The genes in mitochondrial DNA are passed on only by the mother, to all of her children, sons and daughters. Fathers never pass it on. These disorders can turn up in both sexes, in every generation, but how severe they are varies, because of the bottleneck effect: symptoms only appear once the mutated mitochondrial DNA passes a threshold, often sixty to ninety per cent. One example is LHON, which damages the optic nerve.

The traps. Codominance and incomplete dominance are the pair she uses to catch you. Codominance: both shown in full, like A B blood. Incomplete: a blend. Second: her decks put sickle cell under incomplete dominance, so write it her way. Third: mitochondrial means from the mother to all her children. Never from the father.

Once more. The four patterns.

Codominance: both alleles are fully expressed, like A B blood, where I A and I B together give group A B. Incomplete dominance: one allele doesn't fully mask the other, so the heterozygote is in between, like sickle cell carriers with both normal and sickle cells. Polygenic: many genes control one trait, giving a continuous range, like height and skin colour. Mitochondrial: mitochondrial DNA comes only from the mother, to all her children, and severity varies because of the bottleneck effect.